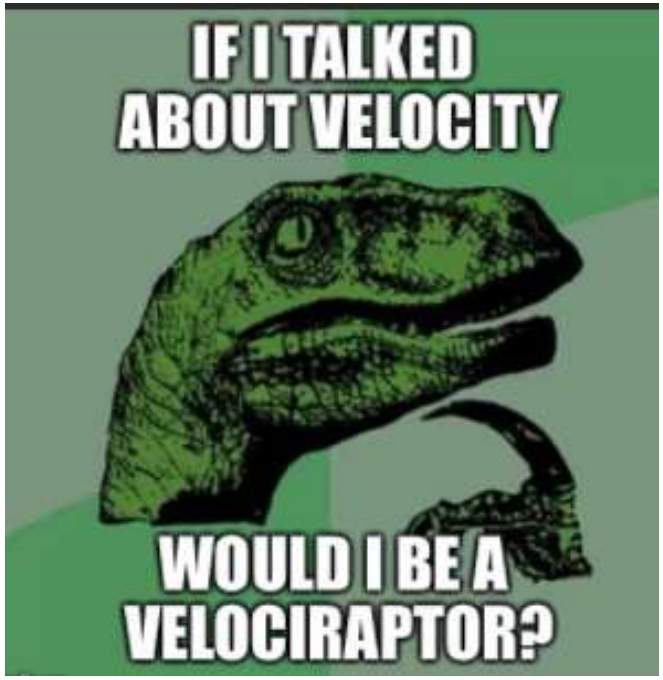
Quiz and practice

Please to prepare a piece of paper to reply, textbook closed (as in exam …)

Question 1



- Considering an arbitrary motion of a particle described by the position vector $\vec{r}(t)$, describe its velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$ in Cartesian coordinates (t is the time) and relationship between them (vector form and Cartesian components).
- Same question about their rotational counterpart: angular position $\theta(t)$, angular velocity vector $\vec{\omega}$ and its z-component ω_z, angular acceleration vector $\vec{\alpha}$ and its z-component α_z (Oz axis is axis of rotation).

Question 1

$$\vec{r}(t) = \begin{Bmatrix} x(t) \\ y(t) \\ z(t) \end{Bmatrix} \qquad \vec{v}(t) = \begin{Bmatrix} v_x(t) \\ v_y(t) \\ v_z(t) \end{Bmatrix} = \begin{Bmatrix} dx(t)/dt \\ dy(t)/dt \\ dz(t)/dt \end{Bmatrix}$$

$$\vec{a}(t) = \begin{Bmatrix} a_x(t) \\ a_y(t) \\ a_z(t) \end{Bmatrix} = \begin{Bmatrix} dv_x(t)/dt \\ dv_y(t)/dt \\ dv_z(t)/dt \end{Bmatrix}$$

Question 1

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt}$$

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt}$$

Question 1

$$\omega_z = \frac{d\theta}{dt}$$

$$\alpha_z(t) = \frac{d\omega_z(t)}{dt} \qquad \vec{\alpha}(t) = \frac{d\vec{\omega}(t)}{dt}$$

Warning: the Cartesian component of a vector could be positive or negative, but the magnitude of a vector (here $\omega = |\vec{\omega}|$, $\alpha = |\vec{\alpha}|$) is always positive

Question 2

Describe the Newton's 2nd law for a rigid body to describe its translational motion and the rotational analogy of the Newton's second law to describe its rotational motion.

Question 2

Newton's second law:

$$m\vec{a} = \vec{F}$$

Mass of the body

Acceleration vector of the center of mass of the body

Net force exerted on the body

Rotational Newton second law:

$$\mathrm{I}\vec{\alpha} = \vec{\tau}$$

Moment of inertia about the axis of rotation of the body

Angular acceleration vector

Net torque exerted on the body about the pivot

Question 3

Describe the gravitational potential energy associated with the weight of a body, the elastic potential energy in a spring-mass system associated with the spring force, and the (translational) kinetic energy of a body. What could change the mechanical energy of a body ?

Question 3

Gravitational potential energy associated with the weight:

$$E_{p,grav} = mgz$$

Mass of the body

Height from the ground (or any arbitrary origin chosen for $z = 0$)

Gravitational acceleration constant: $g = 9.81\ m, s^{-2}$

Question 3

Elastic potential energy associated with the spring force:

$$E_{p,el} = \frac{1}{2}kx^2$$

Distance from equilibrium (we choose $x = 0$ is position of equilibrium)

Spring coefficient,
Or spring force constant

Question 3

Translational Kinetic energy of a body of mass m and velocity v:

$$E_k = \frac{1}{2} m v^2$$

Velocity (magnitude of the velocity vector)

Mass of the body

What could change the mechanical energy of a body ?

$$\Delta E_m = W_{other}$$

Work done by non-conservative forces such as the friction forces

I reminds you: $E_m = E_k + E_p$

Question 4

Describe the maximum magnitude of the static friction force $F_{s,max}$ and the magnitude of the kinetic friction force F_k of a body on a table (coefficient of static friction μ_S, coefficient of kinetic friction μ_k, normal force exerted on the body by the table $\vec{n}$)

Question 4

For the static friction force magnitude:

$$F_s \leq \mu_S n$$

Coefficient of static friction

Magnitude of the normal force

Maximum magnitude of the static friction force:

$$F_{s,max} = \mu_S n$$

Question 4

For the kinetic friction force magnitude:

$$F_k = \mu_k n$$

Coefficient of kinetic friction

Magnitude of the normal force

Question 5

Describe the principle of conservation of (linear) momentum $\vec{p}$ and the principle of conservation of angular momentum $\vec{L}$. For what kind of problem these principles are very useful ?

Question 5

Describe the principle of conservation of (linear) momentum $\vec{p}$

$$\frac{d\vec{p}}{dt} = \vec{F} \quad \Rightarrow \quad \text{If } \vec{F} = \vec{0} \text{ then } \frac{d\vec{p}}{dt} = \vec{0}$$

Momentum is conserved

Net force exerted on a system (external forces)

Question 5

Describe the principle of conservation of angular momentum $\vec{L}$ of a body in rotation:

$$\frac{d\vec{L}}{dt} = \vec{\tau} \quad \Rightarrow \quad \text{If } \vec{\tau} = \vec{0} \text{ then } \frac{d\vec{L}}{dt} = \vec{0}$$

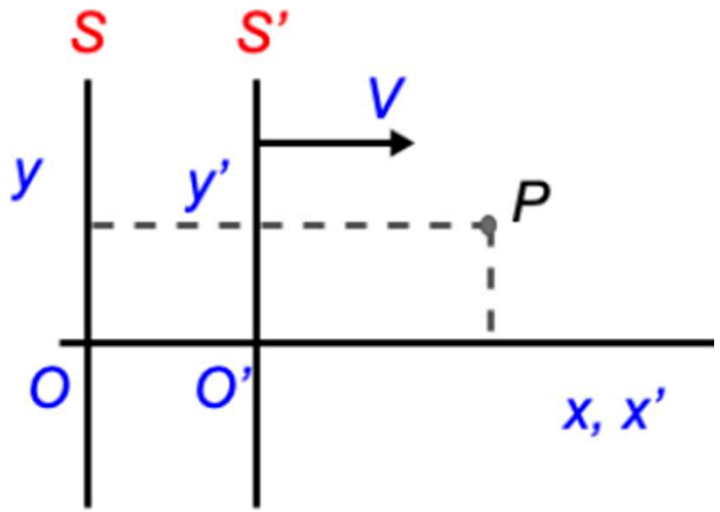
Angular Momentum is conserved

Net torque exerted on a system (external forces) about a point

Net torque and angular momentum are always defined about a point (usually we consider a point of the axis of rotation).

Principle of conservation of momentum is useful to describe collisions when there is not external force exerted on a system of two bodies.

Question 6



An inertial frame of reference S' = $\{O', x', y', z'\}$ has velocity V with respect to inertial reference frame S = $\{O, x, y, z\}$, moving along the +x-direction. Both reference frames coincide at time $t = 0$.

- Describe x' (the x-coordinate of a particle P in S') in terms of x (the x-coordinate of a particle in S), V, time t.
- Describe $v_x' = dx'/dt$ in terms of $v_x = dx/dt$ and V.

Question 6

$$x' = x - Vt$$

$$v_x' = v_x - V$$

Question 7

A spaceship fires a missile at 2000 m/s with respect to the spaceship. The spaceship move with respect to the ground in the same direction at speed 1000 m/s. What is the speed of the missile with respect to the ground ?

$3000\ m/s$

Question 8

What is this work by a force which is constant exerted on a body in straight-line displacement ?

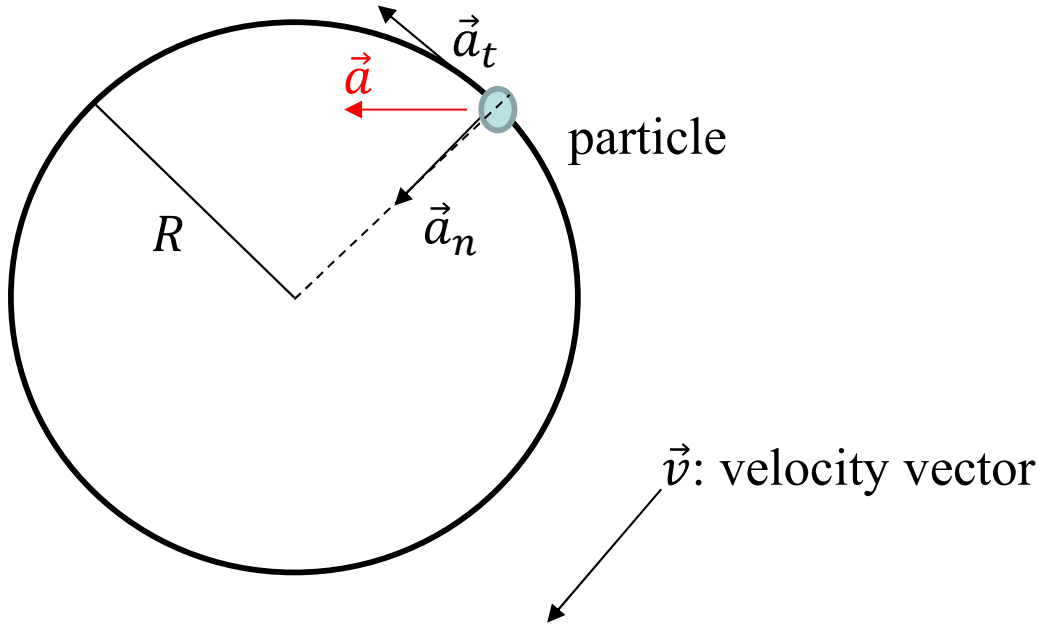
For straight-line displacement $\vec{s}$, the work done by a constant force $\vec{F}$ on the body is:

$$W = \vec{F}.\vec{s}$$

Question 9

A particle has a circular motion of radius R, describe its tangential and normal acceleration, and its angular velocity ω in terms of its velocity v. Same question if the circular motion is uniform circular motion.

Question 9



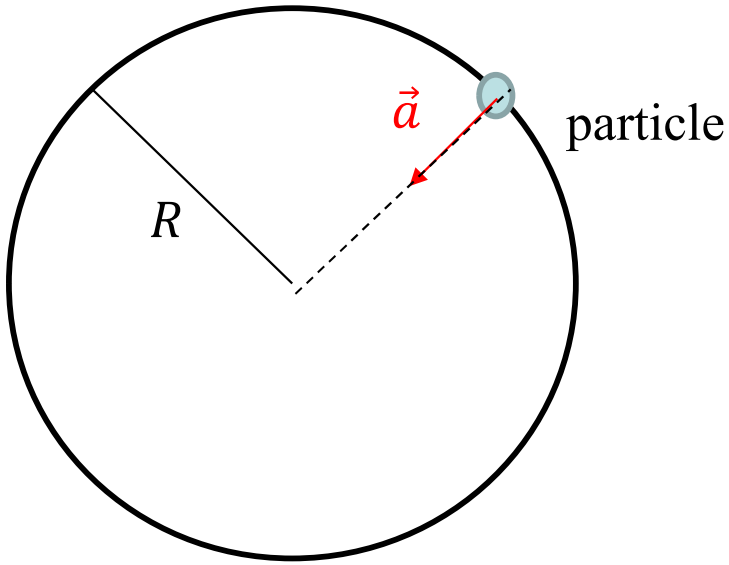
Tangential acceleration: $\vec{a}_t = a_t\hat{t} = \dfrac{dv}{dt}\hat{t}$

Normal acceleration (i.e radial acceleration): $\vec{a}_n = a_n\hat{n} = \dfrac{v^2}{R}\hat{n}$

Angular velocity : $\omega(t) = \dfrac{v(t)}{R}$

$\hat{n}$: unit vector normal to the trajectory
$\hat{t}$: unit vector tangent to the trajectory

Question 9



If the circular motion is uniform, $v = constant$

Tangential acceleration: $\vec{a}_t = a_t\hat{t} = \frac{dv}{dt}\hat{t} = \vec{0}$

Normal acceleration (i.e radial acceleration): $\vec{a}_n = a_n\hat{n} = \frac{v^2}{R}\hat{n} = \vec{a}$

Angular velocity : $\omega(t) = \frac{v}{R} = constant$

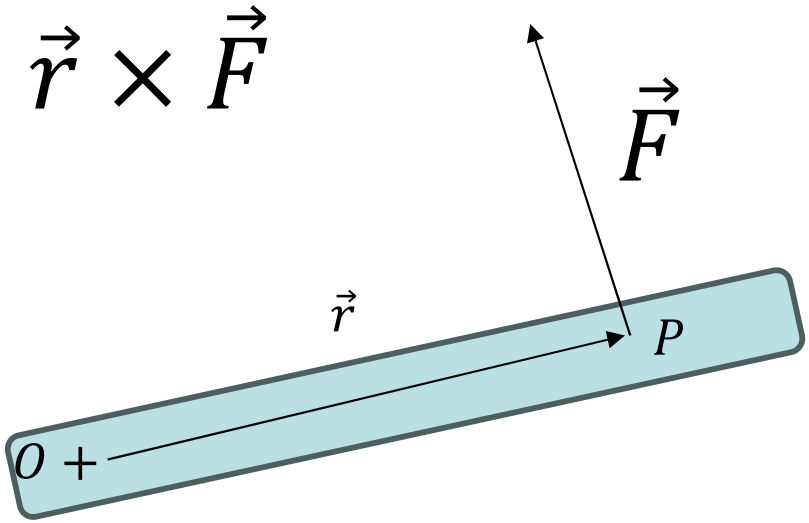
$\hat{n}$: unit vector normal to the trajectory
$\hat{t}$: unit vector tangent to the trajectory

Question 10

- Describe the torque by a force (draw a figure, give the equation) about a point O, applied on a point P.

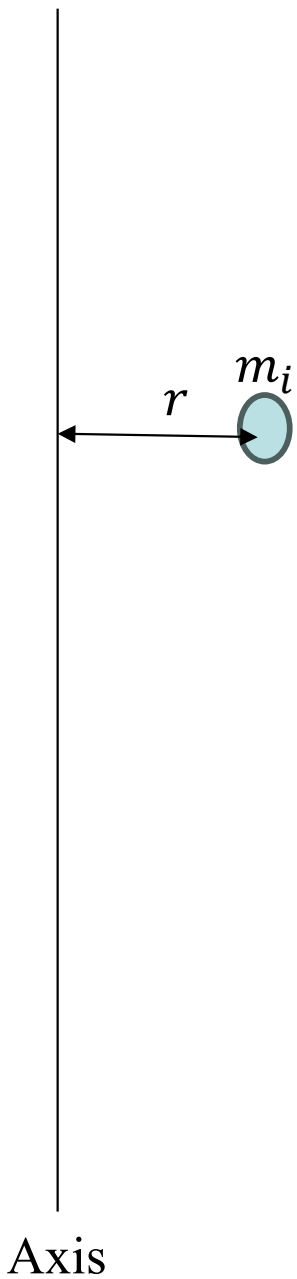
Question 10

$$\vec{\tau} = \vec{r} \times \vec{F}$$



The point O could be any point but we usually consider a point of the axis of rotation of rigid body (same for the angular momentum about a point O).

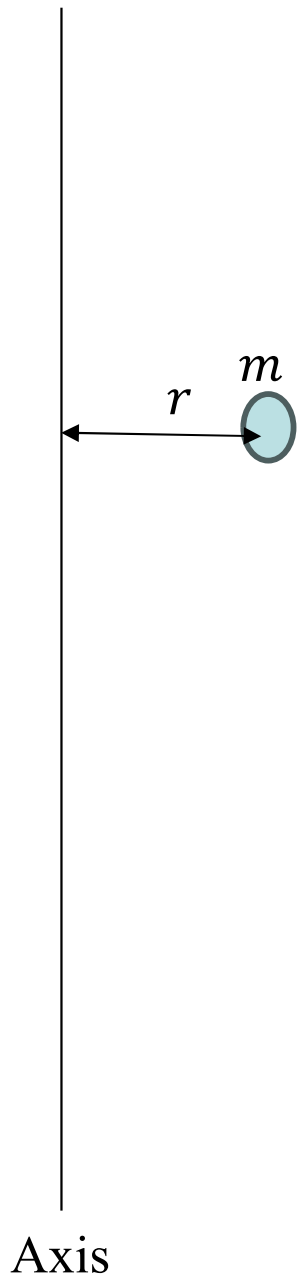
Question 11



Describe the moment of inertia of a particle of mass m_i about an axis.

Describe the rotational kinetic energy of a rigid body in terms of its moment of inertia about axis of rotation and angular velocity ω

Question 11



Moment of inertia of a particle about an axis:

$$I_i = m_i r^2$$

Rotational kinetic energy of a rigid body

$$E_{K,rot} = \frac{1}{2} I \omega^2$$

I: moment of inertia **about the axis of rotation**

Question 12

- A sinusoidal wave (the perturbation is y-displacement) on a string is propagating along the +x-direction. Describe $y(x,t)$ in terms of amplitude A, wave number k, angular frequency ω, phase at origin φ_0.
- The Cartesian system is such the phase at origin is zero. What is $y(x=0,t=0)$?

Question 12

$$y(x,t) = Acos\,(\omega t - kx + \varphi_0)$$

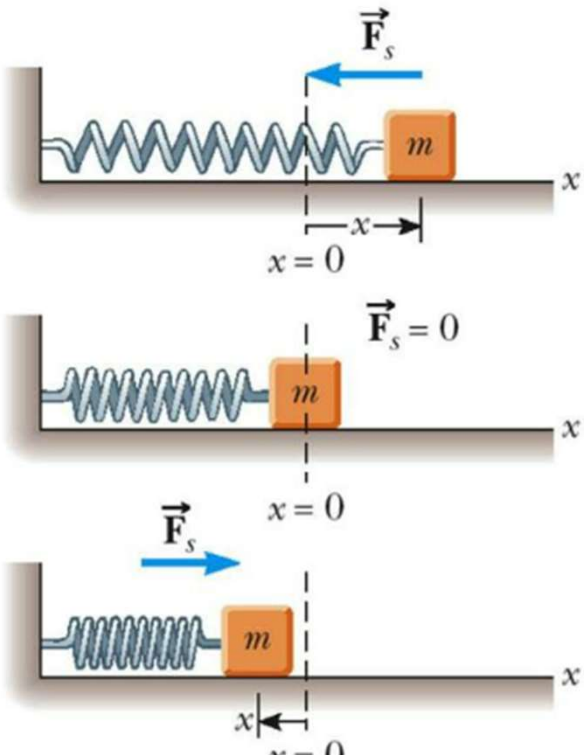
If the Cartesian system is such the phase at origin is zero,

$$y(x,t) = Acos\,(\omega t - kx)$$

$$y(x = 0, t = 0) = A$$

Question 13

- Describe a simple harmonic motion $x(t)$ and the corresponding differential equation for a horizontal spring-mass system. Give the angular frequency of SHM of the frictionless idealized spring mass system (in terms of spring constant k, mass m). Describe the potential energy ($x = 0$ is chosen to be equilibrium position), the kinetic energy and the mechanical energy of the idealized spring mass system.



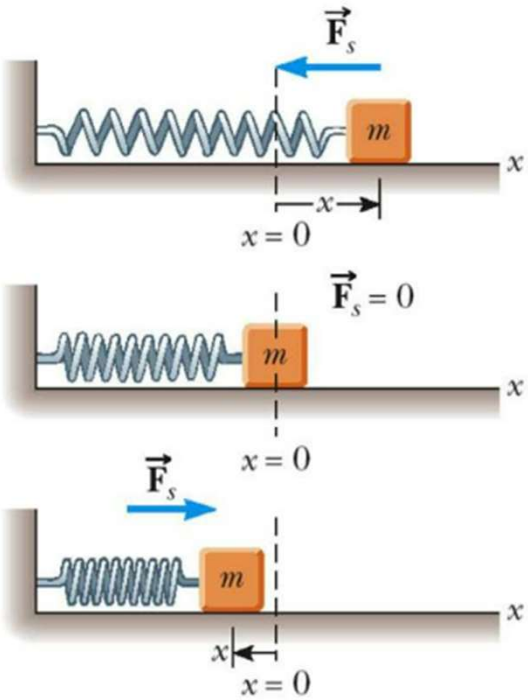
Question 13

$$x(t) = A\cos(\omega t + \varphi_0)$$

Solution of:

$$\frac{d^2x(t)}{dt^2} + \omega^2 x(t) = 0$$

A: amplitude of the SHM
ω: angular frequency of the SHM
φ_0: Phase at origin of the SHM



Angular frequency of the spring-mass system (spring constant k, mass m):

$$\omega = \sqrt{\frac{k}{m}}$$

$$ma_x = -kx \Rightarrow \frac{d^2x(t)}{dt^2} + \frac{k}{m}x(t) = 0$$

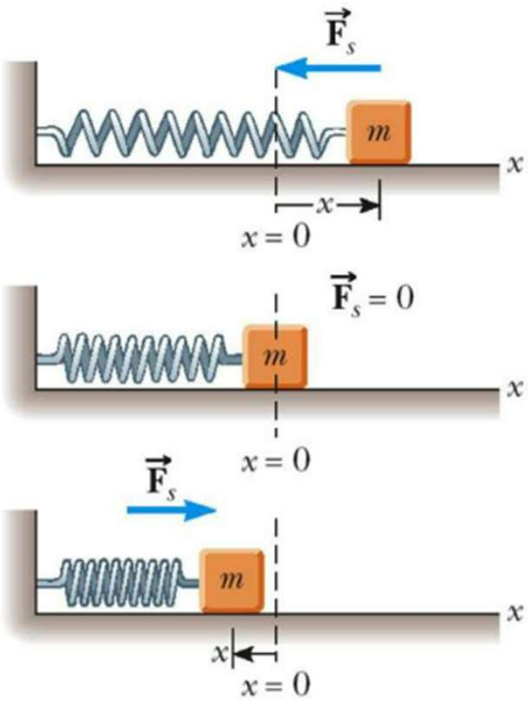
Question 13

Potential energy ($x = 0$ is chosen to be equilibrium position):

$$E_p = \frac{1}{2}kx^2$$

Kinetic energy:

$$E_k = \frac{1}{2}mv^2$$



Mechanical energy (all friction ignored in the case of idealized spring mass system, so the mechanical energy is constant):

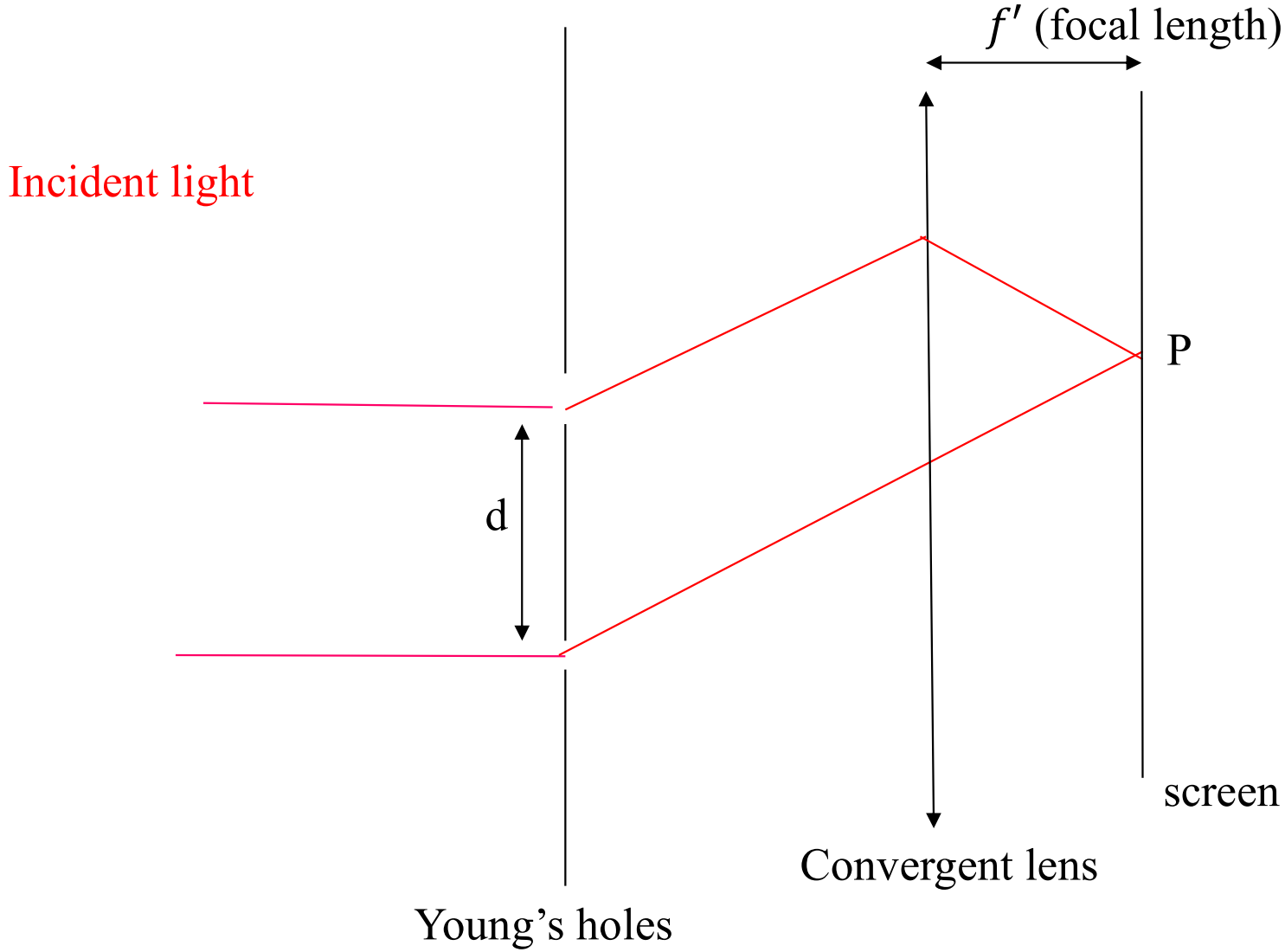
$$E_m = E_k + E_p = \frac{1}{2}kA^2$$

Question 14

- A sinusoidal wave propagates at velocity v . Describe v in terms of angular frequency ω and wave number k and in terms of period T and wavelength λ.

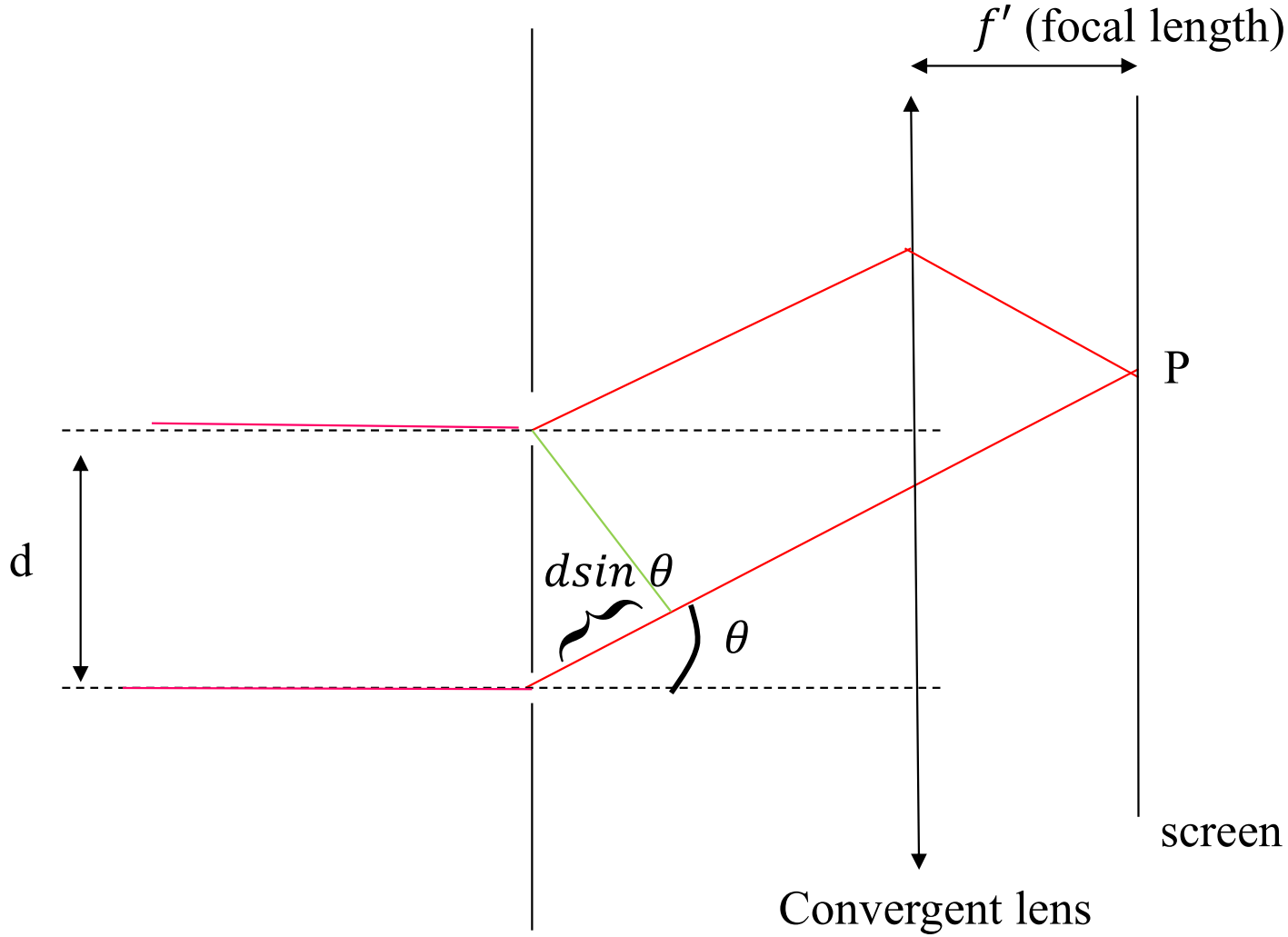
$$v = \frac{\omega}{k} = \frac{2\pi/T}{2\pi/\lambda} \Rightarrow v = \frac{\lambda}{T}$$

Question 15



For the case of Young's experiment, describe a condition of constructive and destructive interference at P between both rays shown at the figure at the screen in a medium of refractive index n.

Question 15



Difference of optical path between both rays:

$$\delta = nd(\sin\theta)$$

n: refractive index of medium of propagation (~1 for air)

Question 15

Condition constructive interferences, the phase difference between both waves $\Delta\varphi$ is such as:

$$\cos\Delta\varphi = 1$$

$$\Delta\varphi = \cdots, -2\pi, 0, 2\pi, \ldots$$

$$\frac{2\pi}{\lambda_0}\delta = \cdots, -2\pi, 0, 2\pi, \ldots$$

$$\delta = \cdots, -\lambda_0, 0, \lambda_0, \ldots \qquad \delta = m\lambda_0$$

$$m = \cdots, -1, 0, 1, \ldots$$

$$nd\sin\theta = \cdots, -\lambda_0, 0, \lambda_0, \ldots$$

δ: optical path difference between both waves.

Question 15

Condition of destructive interferences:

$$\cos\Delta\varphi = -1 \qquad \Delta\varphi = \cdots, -3\pi, -\pi, \pi, 3\pi \ldots$$

$$\frac{2\pi}{\lambda_0}\delta = -3\pi, -\pi, \pi, 3\pi \ldots$$

$$\delta = \cdots, -\frac{3\lambda_0}{2}, -\frac{\lambda_0}{2}, \frac{\lambda_0}{2}, \frac{3\lambda_0}{2} \ldots$$

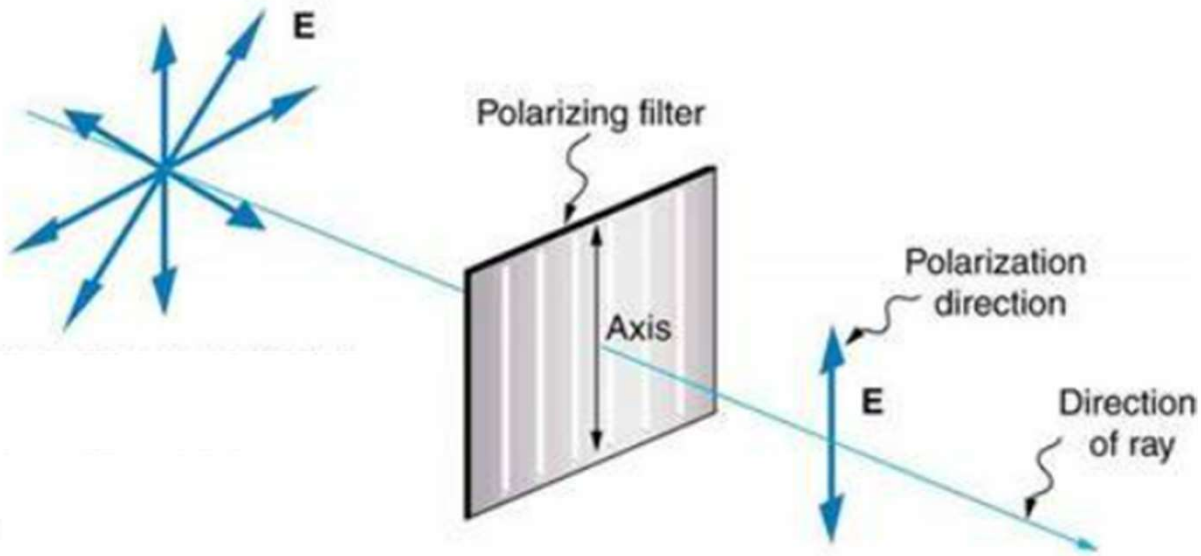
$$\delta = \left(m + \frac{1}{2}\right)\lambda_0 \qquad m = \cdots, -1, 0, 1, \ldots$$

$$nd\sin\theta = \left(m + \frac{1}{2}\right)\lambda_0$$

$$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda$$

Question 16

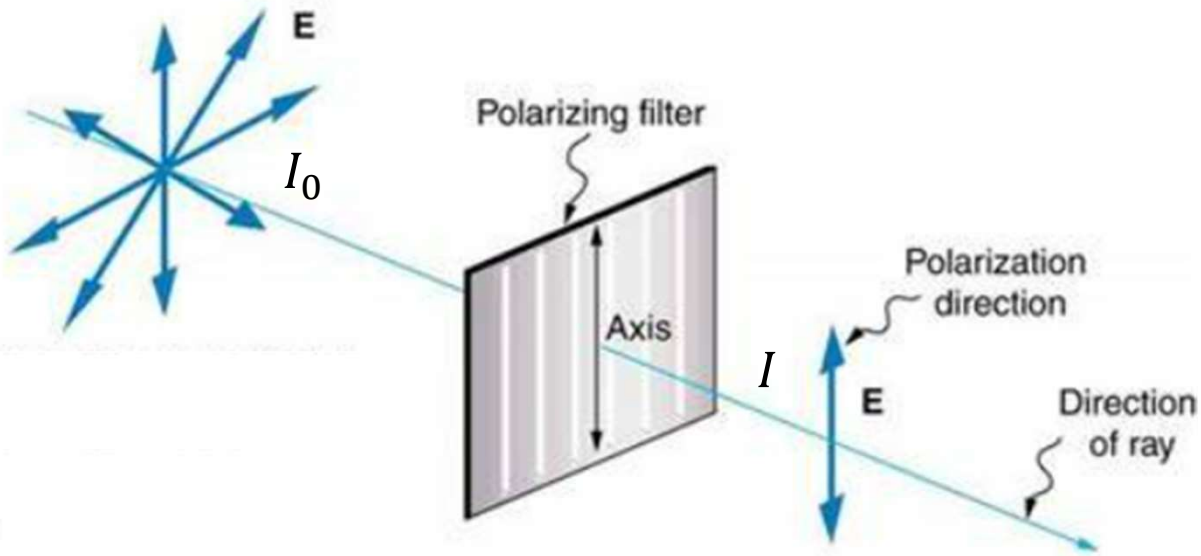
- If unpolarized incident light of intensity I_0 pass through a polarizing filter, what is the intensity I after the polarizer filter, in terms of I_0 ? Now we consider the incident light is polarized linearly. What is the intensity after the polarizing filter in terms of I_0? (draw a figure)



Question 16

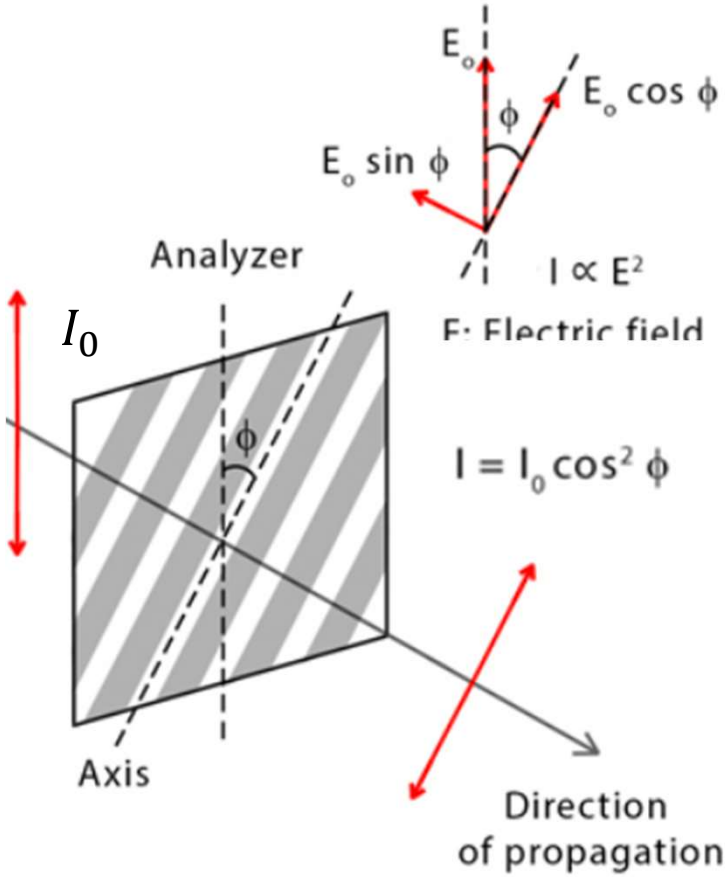
- If unpolarized incident light of intensity I_0 pass through a polarizing filter, what is the intensity I after the polarizer filter, in terms of I_0 ?

$$I = \frac{I_0}{2}$$



Question 16

- Now we consider the incident light is polarized linearly. What is the intensity after the polarizing filter ? (draw a figure)



Question 17

- Describe the relation between:
  - The speed of light v in medium of propagation of refractive index n and its speed in vacuum c,
  - The wavelength of the light λ in a medium of refractive index n and the wavelength in vacuum λ_0.

Question 17

Relation between the speed of light v in medium of propagation of refractive index n and its speed in vacuum c:

$$v = \frac{c}{n}$$

Relation between the wavelength of the light λ in a medium of refractive index n and the wavelength in vacuum λ_0.

$$\lambda = \frac{v}{f} = \frac{c}{nf} = \frac{\lambda_0}{n}$$

Warning: $\lambda = \frac{\lambda_0}{n(\lambda_0)}$

Question 18

- A wave is reflected at the interface between two media of refractive index n_1(medium of incidence) and n_2. For which is condition there is or there isn't half-wave loss ? (Draw a figure)
- What is the meaning of the half-wave loss about the phase of the reflected wave ?
- What is the meaning of the half-wave loss for the optical path and for the path of the reflected wave ?

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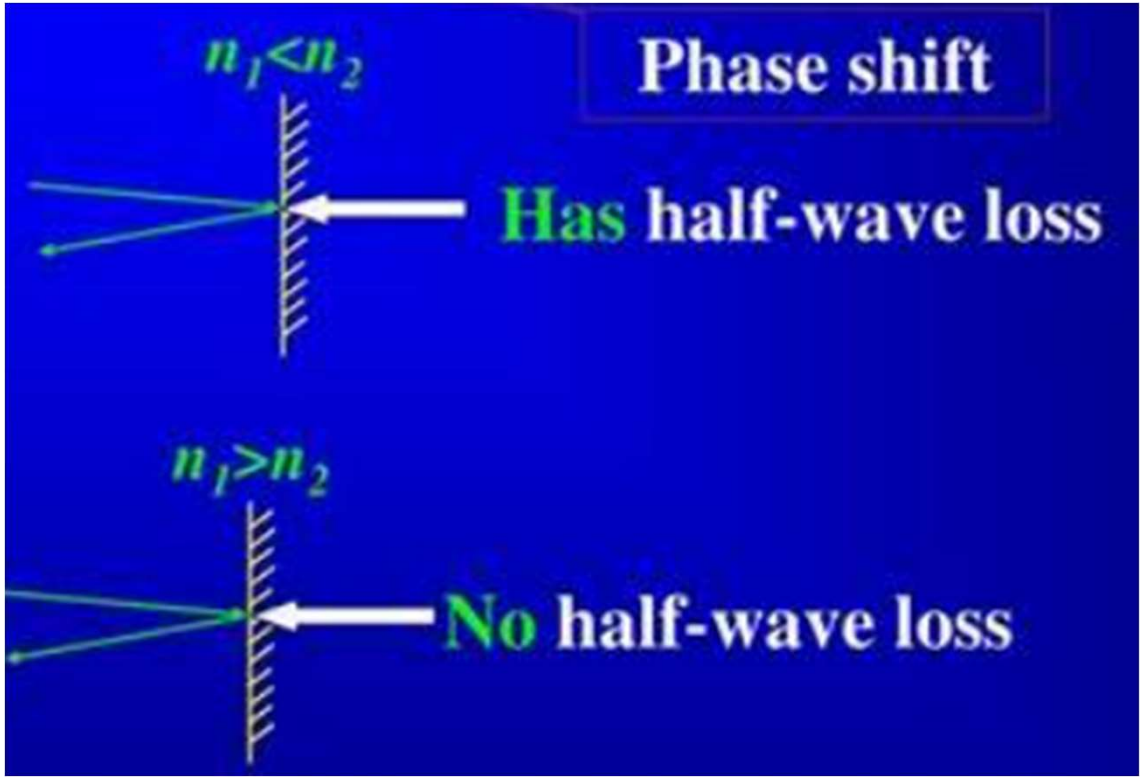

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- What is the meaning of the half-wave loss about the phase of the reflected wave ?
- What is the meaning of the half-wave loss for the optical path and for the path of the reflected wave ?

Half-wave loss corresponds to an extra-phase of π for the reflected wave.

$$\frac{2\pi}{\lambda_0}\delta = \frac{2\pi}{\lambda_0}n_1 l = \pi \quad \Rightarrow \quad \delta = \frac{\lambda_0}{2}$$

Extra-optical path to consider equals half of wavelength in vacuum of the wave

$$l = \frac{\lambda_0}{2n_1} = \frac{\lambda}{2}$$

Extra-path to consider equals half of wavelength in the medium of propagation (of the reflected wave)

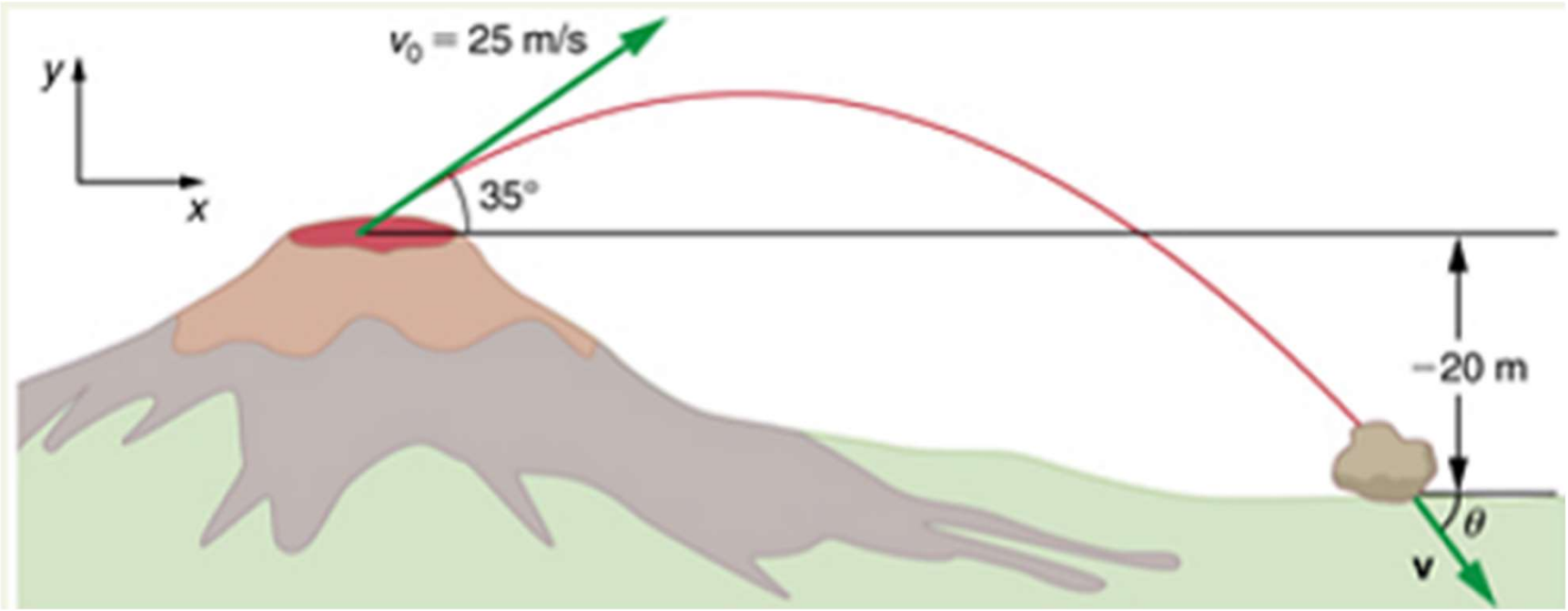
Few exercises for practice

Take care there could something seen in today's quiz/practice could be not in exam, and something could be in exam but not in today's quiz/practice.

Warning: all the questions in the exam **have already been given in homework, quick test (if you did), and ppts …**

Ex. 1

Suppose a large rock is ejected from the volcano with a speed of 25.0 m/s and at an angle 35.0º above the horizontal, as shown in the figure below. The rock strikes the side of the volcano at an altitude 20.0 m lower than its starting point. Calculate the time it takes the rock to follow this path.



Use gravitational acceleration $g = 9.80\ m.s^{-2}$

(I give 5 minutes, it doesn't matter if this time is too short)

Ex. 1

Straight-line motion with constant acceleration:

$$y(t) = y_0 + v_{0,y}t + \frac{1}{2}a_y t^2 = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

The y-acceleration is $a_y = -g = -9.80\ m.s^{-2}$

$$v_{0,y} = v_0 \sin\theta_0 = (25\ m.s^{-1}) \sin 35.0° = 14.3\ m.s^{-1}$$

The equation to solve is the time at which the rock as altitude 20 m lower than its starting point is:

$$-20.0\ m = (14.3\ m.s^{-1})t - (4.90\ m.s^{-2})t^2$$

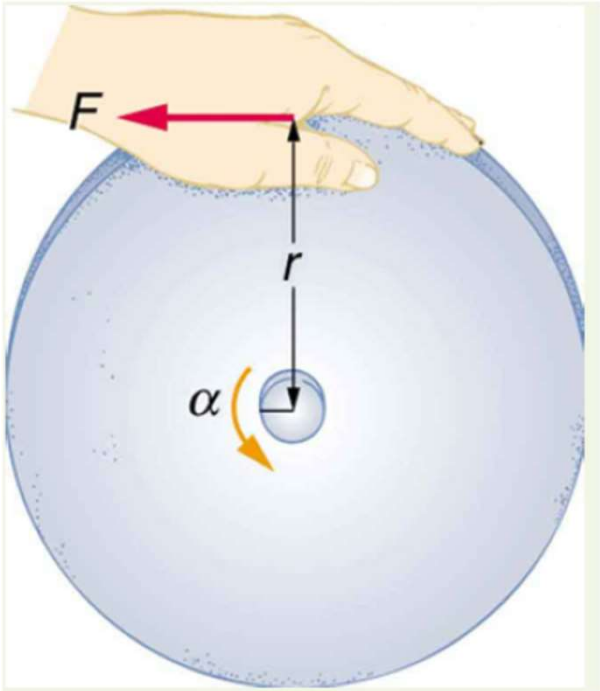
Which is the quadratic equation $at^2 + bt + c = 0$ with $a = 4.90$, $b = -14.3$, $c = -20.0$

Two solutions: $t_1 = 3.96$ s $\quad t_2 = -1.03$ s

Only the positive solution is meaningful $\Rightarrow$ $t = 3.96$ s

Ex. 2

- Consider a person who spins a large grindstone by placing her hand on its edge and exerting a force through part of a revolution as shown below. (a) How much work (unit: J) is done if she exerts a force of 200 N through a rotation of 1.00 rad? The force is kept perpendicular to the grindstone's 0.320-m radius at the point of application, and the effects of friction are negligible. (b) What is the final angular velocity (unit: $rad.s^{-1}$) if the grindstone has a mass of 85.0 kg? (c) What is the final rotational kinetic energy (unit: J)?



The rotational work W corresponding to angular displacement θ is $W = \tau\theta$ where τ is the torque exerted.

Moment of inertia of the disk (mass M, radius R) about axis of rotation:

$$I = \frac{1}{2}MR^2$$

Ex. 2

a)

The net rotational work W corresponding to angular displacement θ is $W = \tau_{net}\theta$ where τ_{net} is the net torque on the disk.

$$W = rF\Delta\theta = (0.320\ m)(200\ N)(1.00\ rad) = 64.0\ J$$

$$I = \frac{1}{2}MR^2 = 4.352\ kg.m^2$$

$$\tau_{net} = I\alpha = rF \quad \Rightarrow \quad \alpha = \frac{rF}{I} = \frac{(0.320\ m)(200\ N)}{4.352\ kg.m^2} = 14.7\ rad.s^{-2}$$

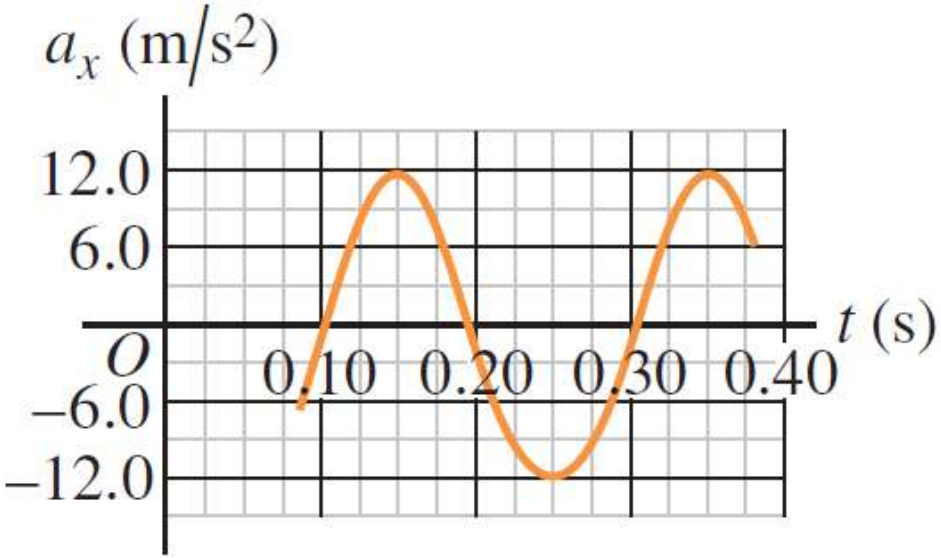
$$\omega^2 = \omega_0^2 + 2\alpha\theta = 2\alpha\theta \quad \Rightarrow \quad \omega = (2\alpha\theta)^{1/2}$$

$$\omega = 5.42\ rad.s^{-1}$$

c) $E_{k,rot} = \frac{1}{2}I\omega^2 = 64.0\ J$

Ex. 3

On a frictionless, horizontal air track, a glider of mass m oscillates at the end of an ideal spring of force constant (also named spring coefficient) $k = 250\ N.m^{-1}$. The graph shows the acceleration of the glider as a function of time. (a) Describe the acceleration of the glider in respect with its x-coordinate $x(t)$, k, m ($x = 0$ is the position of equilibrium, and then in respect with the amplitude A, the angular velocity ω and phase at origin ϕ of the simple harmonic motion describing x(t). (b) Describe the mass of the glider in terms of the spring coefficient k and T the period of the simple harmonic motion of the glider; (c) The mass of the glider is $m \approx 0.250\ kg$. Calculate the maximum displacement of the glider from the equilibrium point (unit: m); (d) Calculate the maximum force the spring exerts on the glider (unit: N).



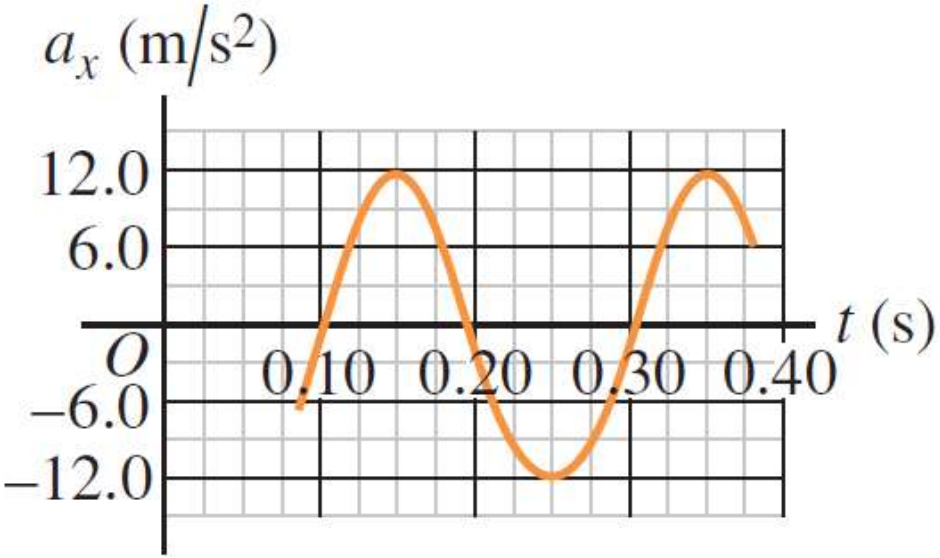
Ex. 3

a)

$$ma_x = -kx$$

$$a_x(t) = -\frac{k}{m}x(t)$$

$$a_x(t) = -\frac{k}{m}Acos(\omega t + \phi)$$



Ex. 3

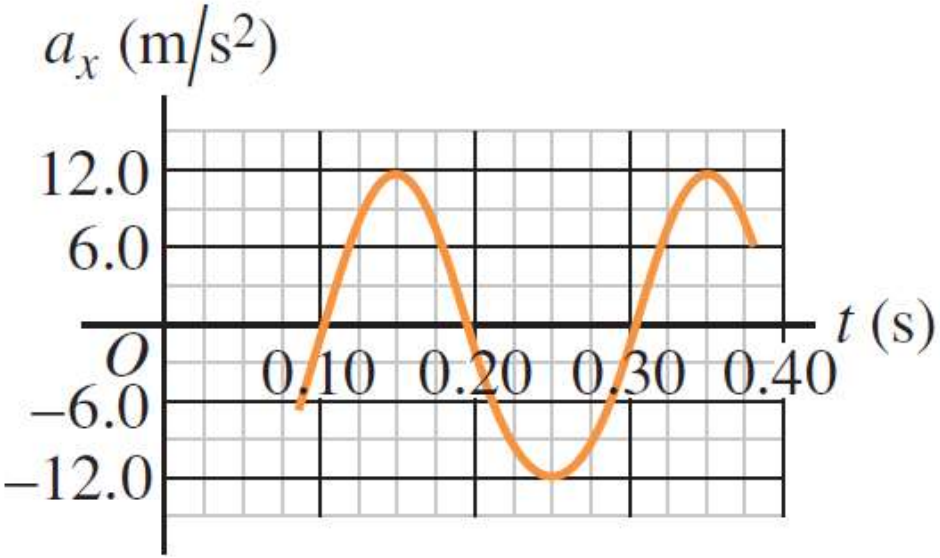
b)

The acceleration oscillate at the same angular velocity as $x(t)$, i.e. $\omega = \sqrt{k/m}$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{m/k}$$

We obtain:

$$m = k\left(\frac{T}{2\pi}\right)^2$$



Ex. 3

c)

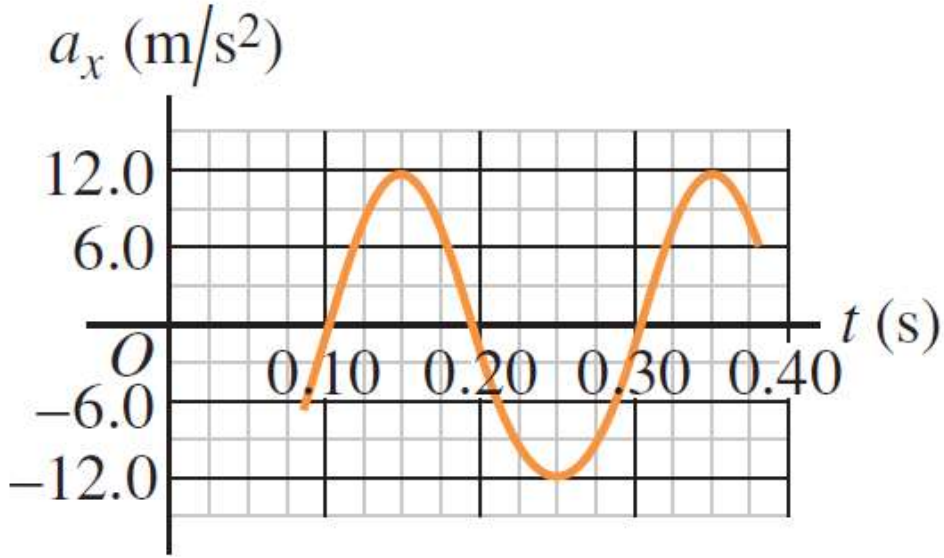
$$a_x(t) = -\frac{k}{m}x(t) = -\frac{k}{m}Acos(\omega t + \phi)$$

$$a_{x,max} = -\frac{k}{m}x_{max} = -\frac{k}{m}A$$

The maximum acceleration is:

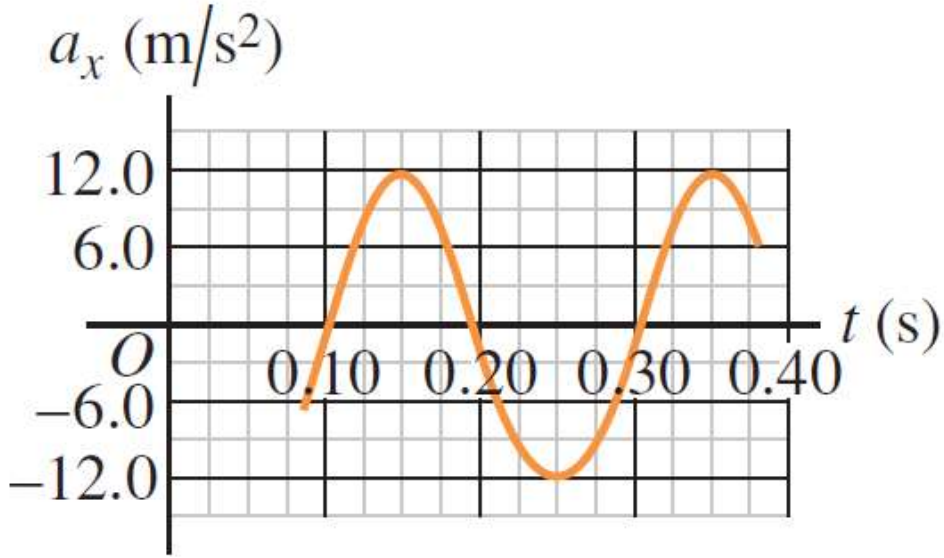
$$a_{max} = \frac{k}{m}A$$

$$A = x_{max} = \frac{ma_{max}}{k} = \frac{(0.250\ kg)(12.0\ m.s^{-2})}{250\ N.m^{-1}} = 0.0120\ m$$



Ex. 3

d) $F_{max} = kA = ma_{max}$
$= (0.250\ kg)(12.0\ m.s^{-2}) = 3.00\ N$



Ex. 4

A certain transverse wave is described by

$$y(x,t) = (6.50\ mm)\cos 2\pi\left(\frac{t}{0.0360\ s} - \frac{x}{28.0\ cm}\right)$$

Determine the wave's (a) amplitude; (b) wavelength; (c) frequency (d) speed of propagation; (e) direction of propagation(you don't have to use SI units, you can choose what is the most convenient according to what you see).

Ex. 4

The wave is described by the equation $y(x,t) = Acos\ (\omega t - kx)$

(a)

The amplitude of the wave is: $A = 6.50\ mm$

(b)

The wave number $k = 2\pi/\lambda$, thus the wavelength of the wave is: $\lambda = 28.0\ cm$.

(c)

The angular frequency is $\omega = 2\pi f = 2\pi/T$ where $T = 0.0360\ s$ is the period of the wave, thus the frequency is:

$$f = \frac{1}{0.0360\ s} = 27.8\ Hz$$

The unit of the frequency Hz means "s^{-1}".

(d) The velocity of propagation of the wave is:

$$v = \frac{\lambda}{T} = \frac{0.28\ m}{0.0360\ s} = 7.78\ m.s^{-1}$$

(e) The wave propagates toward the +x-direction.

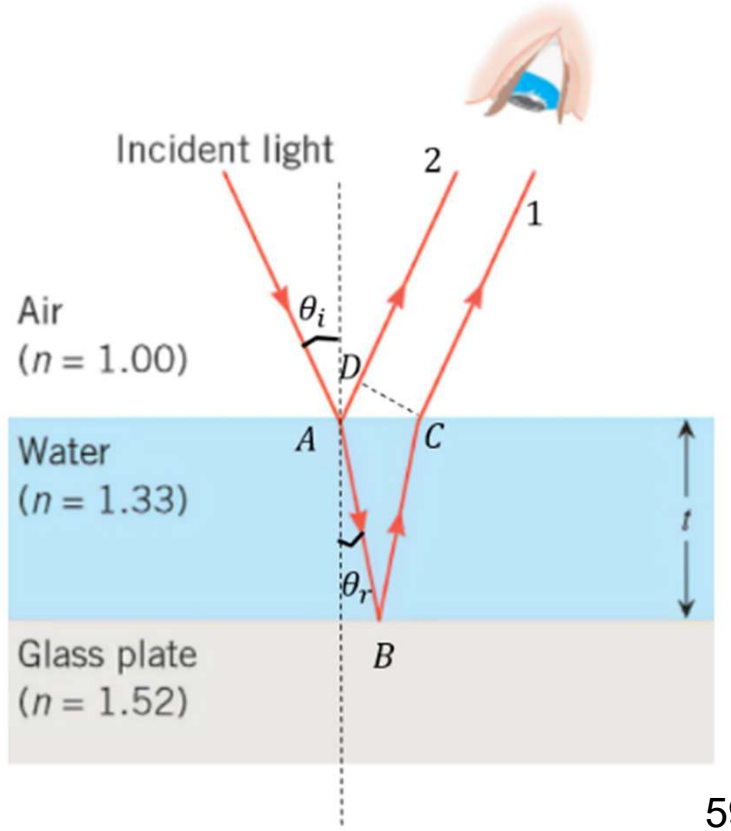
Ex. 5

We consider an almost normal incidence of angle of $\theta_i \approx 0$, such as AD << AB, AD<<BC and $sin\,\theta_i \approx 0$. The incident light has wavelength in vacuum λ_0. The eye has same behavior as a convergent lens or a screen. The thickness of the film of water is t. Both planes of the interfaces air/water and water/glass plate are parallel. The refractive index of water is $n_W = 1.33$ and is $n_G = 1.52$ for the glass. $\theta_r \approx 0$ is the angle of refraction at the interface air/water.

(1) Considering possible halfwave loss, describe the difference of optical path δ between both rays in terms n_W, t (and λ_0 if there are half-wave loss terms to consider)

(2) Describe a condition of constructive interferences between both rays about the thickness t in terms of n_W, λ_0 and a positive integer m.

(3) What is the small thickness of the thin film of water to observe constructive interferences in terms of λ_0 and in terms of λ the wavelength in water ?

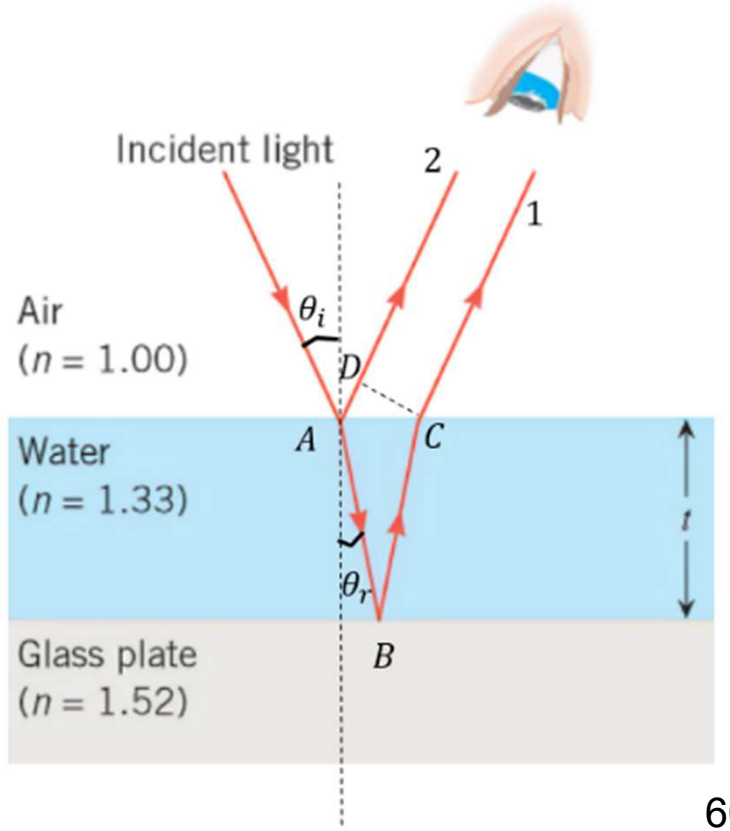


Ex. 5

(1) Half-wave loss: There is here one half-wave loss at the interface air/water for the ray 2, at point A and one half-wave loss for the ray 1, at the interface water/glass at point B, thus $\delta' = \delta$ the optical path difference is the same than if there was no half-wave loss.

Considering almost normal incidence and considering possible half-wave loss, the difference of optical between both rays is:

$$\delta \approx 2n_W t$$



Ex. 5

(2) Both waves interfere if their phase difference $\Delta\varphi$ is such as:

$$\cos\Delta\varphi = 1$$

$$\Delta\varphi = \cdots, -2\pi, 0, \pi, \ldots$$

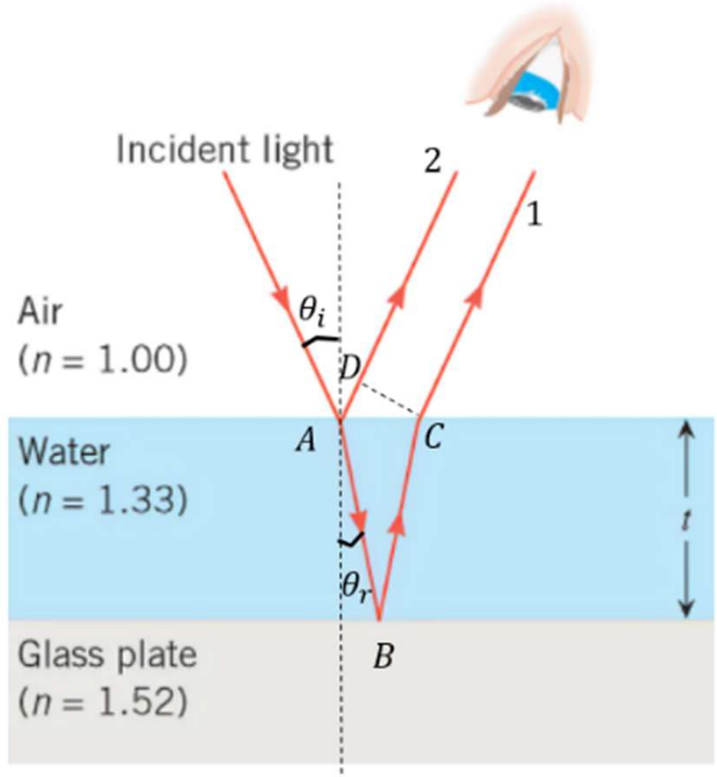
And $\Delta\varphi = \dfrac{2\pi}{\lambda_0}\delta$

$$\delta = 2n_W t = m\lambda_0$$

where m is a positive integer.

The condition of constructive interferences between both waves on t is:

$$t = \frac{m\lambda_0}{2n_W}$$



Warning: m cannot be negative because t, λ_0 and n_W are positive.

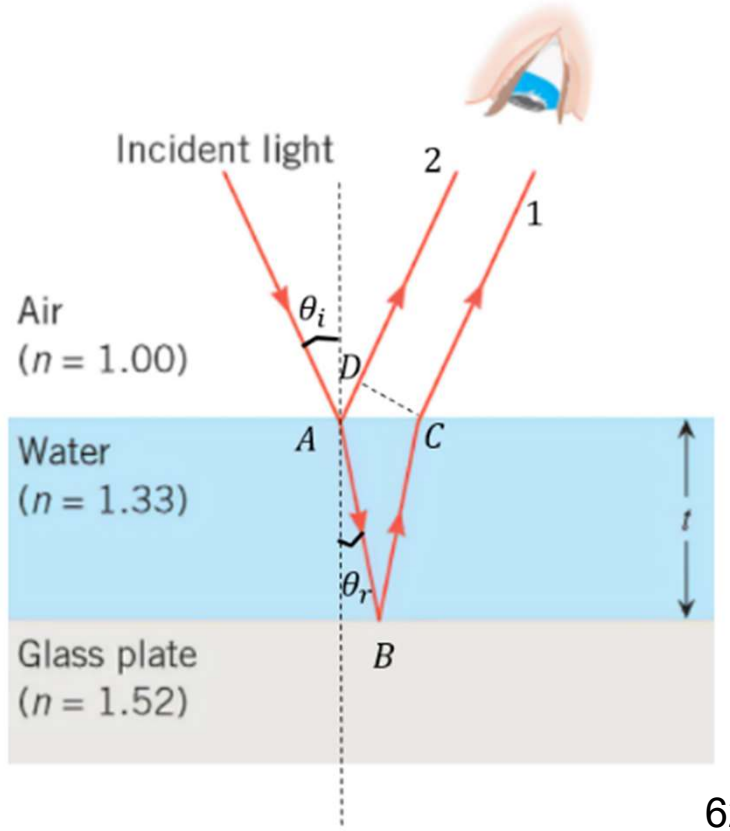
Ex. 5

(3) The condition of constructive interferences between both waves on t is:

$$t = \frac{m\lambda_0}{2n_W}$$

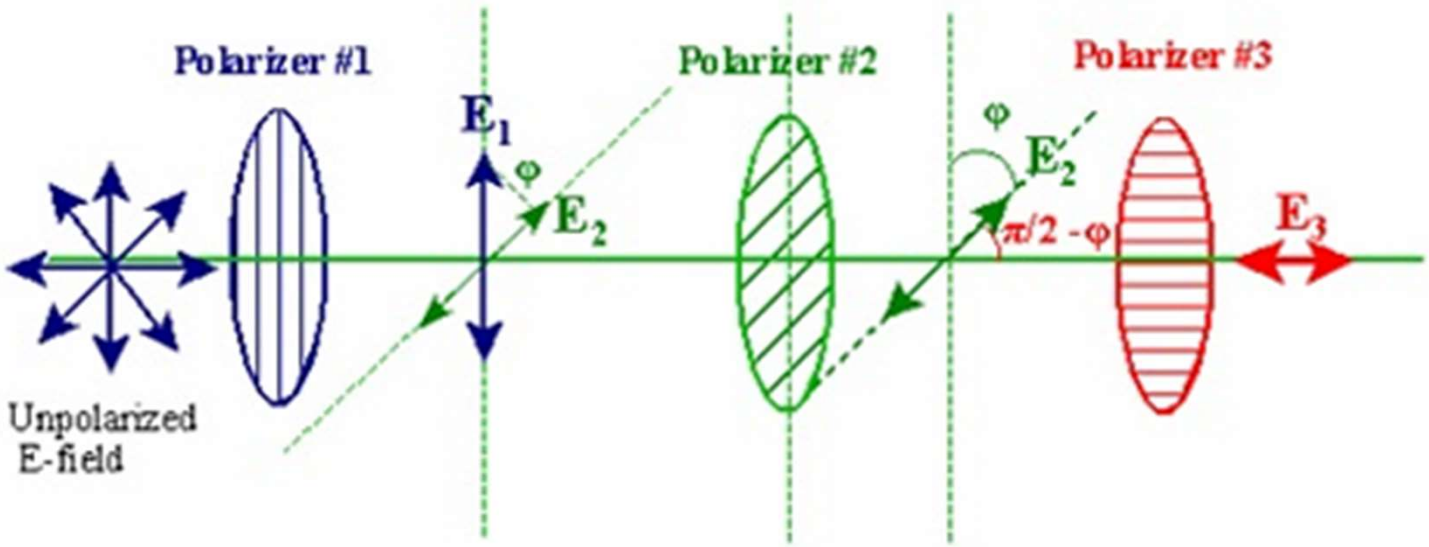
The minimum thickness t to observe constructive interferences is:

$$t_{min} = \frac{\lambda_0}{2n_W} = \frac{\lambda}{2}$$



Ex. 6

Incident unpolarized light pass through 3 polarizing filters as shown in the figure. The first and last polarization axes are oriented at 90° with respect to each other. Describe the intensity I_3 after the last polarized in terms of intensity I_1 after the first polarizer and the angle φ shown in the figure.



Help: $\sin 2\theta = 2\sin\theta\cos\theta$

Ex. 6

After the second polarizer, the intensity of the light is:

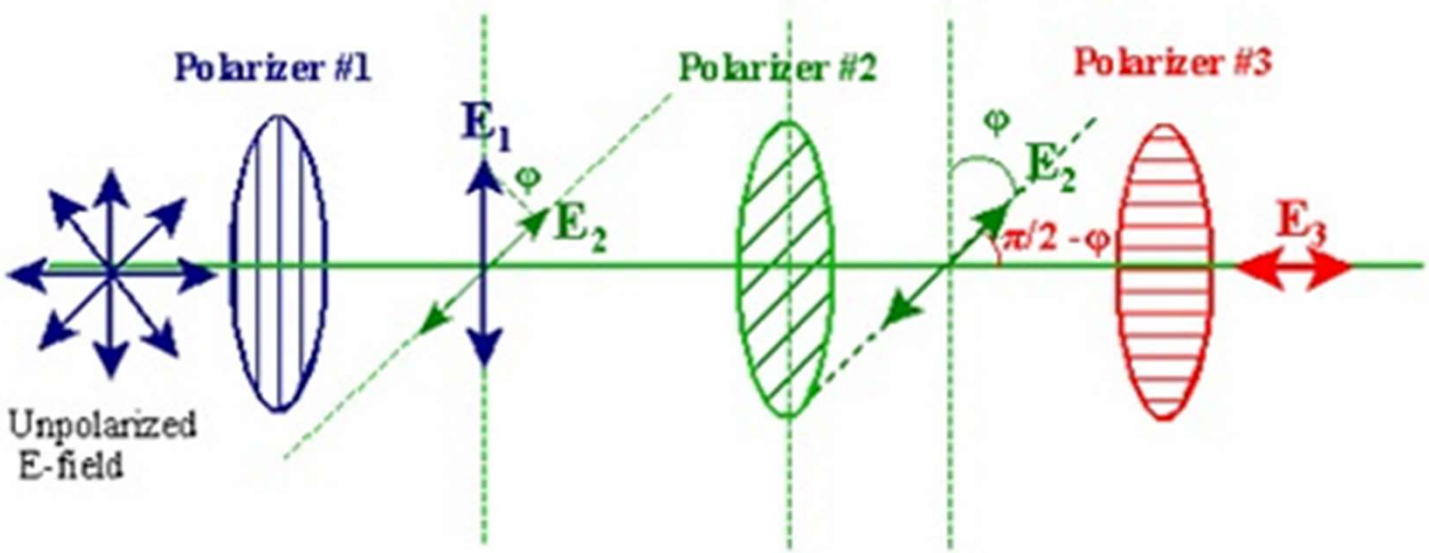
$$I_2 = I_1 \cos^2 \phi$$

After the third polarizer, the intensity of the light is:

$$I_3 = I_2 \cos^2\left(\frac{\pi}{2} - \phi\right) = I_1 \cos^2 \phi \cos^2\left(\frac{\pi}{2} - \phi\right)$$

$$\sin 2\theta = 2 \sin\theta \cos\theta \quad \Rightarrow \quad I_3 = \frac{I_1}{4} \sin^2(2\phi)$$

$$\cos\left(\frac{\pi}{2} - \phi\right) = \sin\phi$$



- Fin du ppt